

WindOps Copilot - Turbine Report

Turbine ID: WTG-002 | Correlation ID: corr-b1fe29e8b81a
Report Generated: 2026-06-26T13:25:04.723008

TURBINE HEALTH STATUS

65.0% Health

Live Sensor Telemetry

Sensor Parameter	Value	Unit
Wind Speed	10.00	m/s
Rotor Speed	12.50	RPM
Blade Temperature	22.00	°C
Generator Temperature	84.50	°C
Vibration	0.11	mm/s
Power Output	0.80	MW
Operational Status	maintenance	

Active Fault Alerts

Parameter	Value	Threshold	Severity
Generator Temp	84.50	80.00	CRITICAL

Diagnostic Root Cause Analysis

Diagnosed Cause:	Cooling Subsystem Blockage
Confidence Level:	90.0%
Technical Analysis:	Generator temperature is critically high (84.5°C) despite low power output (0.80 MW). This indicates the heat-exchange cooling pump or radiators have failed or are blocked.

Operational Action Recommendations

Recommended Action	Description	Risk	Production Loss	Confidence
Curtail Power Output (50% Load)	De-rate the turbine capacity to reduce thermal load and allow generator/pitch systems to cool down while continuing partial production.	LOW	moderate	90%

Schedule Urgent Maintenance	Deploy technician team within 48 hours to clean cooling manifolds and verify pitching gear actuators.	MEDIUM	low	80%
Continue Normal Production	Maintain status quo. (HIGH RISK: temperature levels may continue escalating, risking permanent stator windings insulation damage).	HIGH	none	20%

Note: This report is generated by an autonomous multi-agent system. final actions must be reviewed and authorized by a human operator.